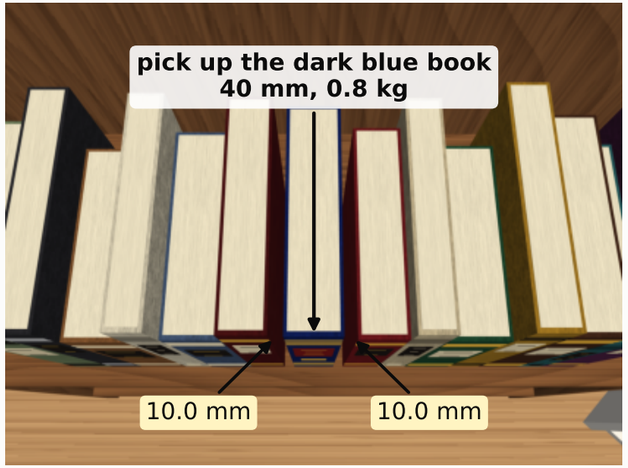
What each column of the results table tests (one test case per suite, seen from above the shelf; the gripper enters from the bottom)

Nominal (simulator scene)

dark-blue 40 mm book, 10 mm gaps, 0.8 kg;
only the start pose changes

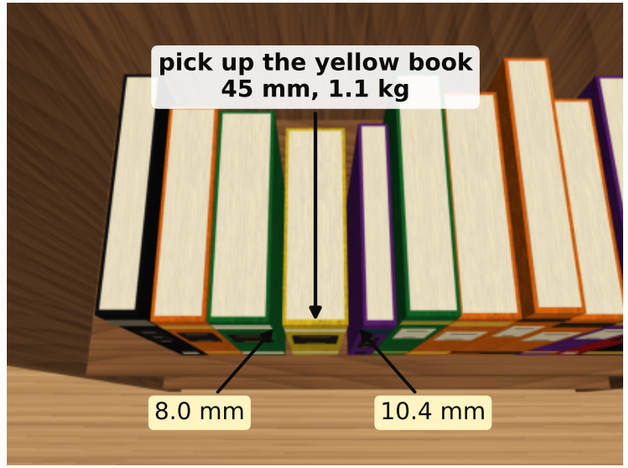


air left beside the 3.5 mm blades: 6.5 mm | 6.5 mm

- π0.5 fine-tuned: 50/50
- scripted expert: 50/50
- π0.5-DROID zero-shot: 0/20

Randomized layouts

any 35-45 mm book, 8-12 mm gaps, 0.4-1.2 kg,
8 colours, sometimes a 2nd gapped book

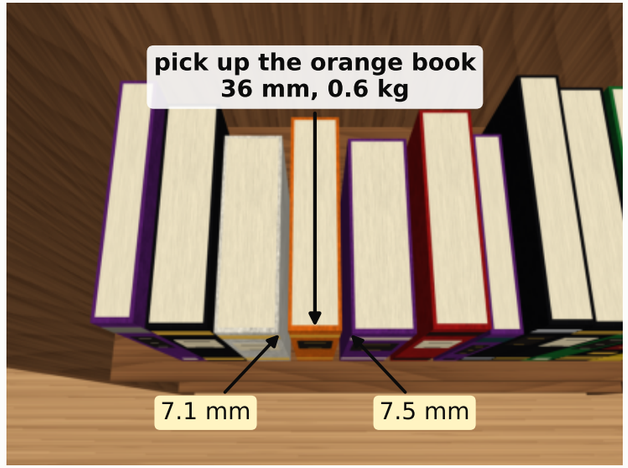


air left beside the 3.5 mm blades: 4.5 mm | 6.9 mm

- π0.5 fine-tuned: 36/50
- scripted expert: 50/50
- π0.5-DROID zero-shot: 0/20

Narrow gaps

6.5-7.5 mm gaps, narrower than
any demonstration

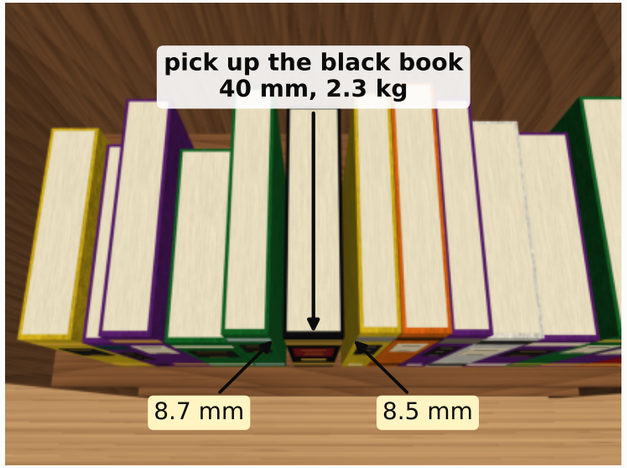


air left beside the 3.5 mm blades: 3.6 mm | 4.0 mm

- π0.5 fine-tuned: 22/30
- scripted expert: 30/30

Heavy + slippery

1.6-2.4 kg book, skin friction 0.5
instead of 1.0



air left beside the 3.5 mm blades: 5.2 mm | 5.0 mm

- π0.5 fine-tuned: 13/30
- scripted expert: 17/30